# J. NITHIN

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## SUMMARY

B.E. ECE student skilled in Embedded C and ESP32/IoT development. Built an automated multi-sensor waste segregation system achieving 92% classification accuracy and an ESP32 GPS telematics tracker streaming live telemetry to ThingSpeak at 5-second intervals. Seeking an entry-level Embedded Systems or IoT Engineer role to build real-time hardware-software solutions.

## EDUCATION

**VV COLLEGE OF ENGINEERING**, Thisaiyanvilai — Expected May 2027

*B.E. Electronics and Communication Engineering*

CGPA: 7.63 / 10

**Higher Secondary (HSC)** — 2023

Percentage: 87.6%

**Secondary School Leaving Certificate (SSLC)** — 2021

ALL PASS (COVID19)

## INTERNSHIP

**IoT Intern** — 12 June 2026 – 03 July 2026

*Malar Software Solutions, Tirunelveli*

- Completed an internship on Internet of Things (IoT). Gained practical exposure to IoT protocols, sensor interfacing, wireless data transmission, and real-time cloud monitoring applications.

## PROJECTS

**Garbage Segregator – Team Leader**

*Academic Hardware & Embedded Project*

- Designed an automated waste segregation system classifying Wet, Dry, and Metal waste with **92% detection accuracy** using IR, inductive proximity, and rain drop sensors.
- Implemented microcontroller-based servo motor actuation achieving rapid bin allocation under **1.8 seconds per item**.
- **Technologies:** Arduino, IR Sensor, Proximity Sensor, Rain Drop Sensor, Servo Motor.

**Smart Micro-Mobility Management**

*IoT Fleet Tracking & Telemetry*

- Engineered an IoT telematics tracker using **ESP32** and GPS module for live fleet location tracking within a **2.5-meter precision radius**.
- Integrated battery health monitoring and streamed telemetry data via **MQTT/Wi-Fi** to **ThingSpeak** with a **5-second update interval**.
- **Technologies:** ESP32, GPS Module, Voltage/Current Sensors, MQTT, ThingSpeak, Arduino IDE.

## TECHNICAL SKILLS

| | | |
|---|---|---|
| **Programming & Embedded** | : | Embedded C (Sensor Interfacing), Python (Basics), Arduino IDE |
| **Microcontrollers** | : | Arduino Uno, ESP32 |
| **Sensors & Modules** | : | GPS Module, IR Sensor, Proximity Sensor, Rain Drop Sensor, LM35, DHT11 |
| **Communication Protocols** | : | UART, SPI, I²C, MQTT |
| **Peripherals** | : | ADC, PWM, LCD, Servo Motor |
| **IoT Platforms** | : | ThingSpeak, ThingsBoard (Basics), Blynk (Basics) |
| **Tools & Platforms** | : | Wokwi, Tinkercad, EasyEDA |
| **VLSI / Digital Design** | : | Verilog HDL (Basics), Digital Circuits, Logic Design |

## ELECTRICAL & ELECTRONICS

- AC/DC fundamentals, network theorems, and circuit analysis
- Single-phase and three-phase power supply fundamentals
- Electrical measurements and testing using a digital multimeter
- Relays, contactors, MCBs, fuses, grounding, and electrical safety standards

## CERTIFICATIONS

- **Embedded Systems Course** – Simplilearn SkillUp — September 2026
- **Internet of Things (IoT)** – Malar Software Solutions — July 2026

## LANGUAGES

- English – Professional Working Proficiency
- Tamil – Native